

Module - III

Statistics for Engineers Correlation and regression

Correlation and Regression - Rank Correlation - Partial and Multiple Correlation - Multiple regression.

Correlation Coefficient

$$\text{Var}(X) = E\{X - E(X)\}^2$$

$$\text{COV}(X, Y) = E\{[X - E(X)][Y - E(Y)]\}$$

{also known as the product moment of X and Y}

$$P(X, Y)$$

$$\frac{P(X, Y)}{\sigma_X \sigma_Y}$$

is a measure of intensity of linear relationship between X and Y and is

Known as Karl-Pearson's product moment correlation coefficient or simply correlation coefficient between X and Y. denoted by $r(X, Y)$ or r_{XY} or simply r .

$$r_{XY} = \frac{E\{[X - E(X)][Y - E(Y)]\}}{\sqrt{E\{X - E(X)\}^2 E\{Y - E(Y)\}^2}}$$

$$= \frac{E(XY) - E(X)E(Y)}{\sqrt{\{E(X^2) - E^2(X)\} \{E(Y^2) - E^2(Y)\}}}$$

$$= \frac{\frac{1}{n} \sum x_i y_i - \frac{1}{n} \sum x_i \cdot \frac{1}{n} \sum y_i}{\sqrt{\left\{ \frac{1}{n} \sum x_i^2 - \left(\frac{1}{n} \sum x_i \right)^2 \right\} \left\{ \frac{1}{n} \sum y_i^2 - \left(\frac{1}{n} \sum y_i \right)^2 \right\}}}$$

$$r_{xy} = \frac{n \sum xy - \sum x \cdot \sum y}{\sqrt{\{n \sum x^2 - (\sum x)^2\} \{n \sum y^2 - (\sum y)^2\}}}$$

Properties of Correlation coefficient

1.) $-1 \leq r_{xy} \leq 1$ or $|\text{COV}(X, Y)| \leq \sigma_x \cdot \sigma_y$

* When $0 < r_{xy} < 1$, the correlation b/w X & Y is said to be +ve or direct.

* When $-1 \leq r_{xy} \leq 0$, -ve or inverse.

* When $-1 \leq r_{xy} \leq -0.5$ or $0.5 \leq r_{xy} \leq 1$

the correlation is assumed to be high,
other assumed to be poor.

2) Correlation coefficient is independent of change of origin and scale.

Example +ve correlation $\rightarrow$ income and expenditure
-ve $\rightarrow$ volume & pressure of a perfect gas

Ex Find the correlation coefficient for the following variables

X	Y	X ²	Y ²	XY
65	67	4225	4484	4355
66	68	4306	4624	4488
67	65	4489	4225	4305
67	68	4489	4629	4556
68	72	4624	5784	4896
69	72	4761	5784	4968
70	69	4900	4761	4830
72	71	5784	5041	5112
$\Sigma X = 554$	$\Sigma Y = 552$	$\Sigma X^2 = 37028$	$\Sigma Y^2 = 38132$	$\Sigma XY = 37520$

$$r_{xy} = \frac{8 \times 37520 - 554 \times 552}{\sqrt{\{8 \times 37028 - (554)^2\} \{8 \times 38132 - (552)^2\}}}$$

$$= 0.6032$$

Alternate way

X	Y	$x = X - 69$	$y = Y - 69$	x^2	y^2	xy
65	67	-3	-2	9	4	6
66	68	-2	-1	4	1	2
67	65	-1	-4	1	16	4
67	68	-1	-1	1	1	1
68	72	0	+3	0	9	0
69	72	1	3	1	9	3
70	69	2	0	4	0	0
72	71	4	2	16	4	8
		$\Sigma x = 0$	$\Sigma y = 0$	$\Sigma x^2 = 36$	$\Sigma y^2 = 44$	$\Sigma xy = 24$

$$\begin{aligned} \text{COV}(x, y) &= \frac{\Sigma xy}{n} - \frac{\Sigma x}{n} \cdot \frac{\Sigma y}{n} \\ &= \frac{24}{8} - 0 = 3 \end{aligned}$$

$$\sigma_x = \sqrt{\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2} = \sqrt{\frac{36}{8} - 0} = 2.121$$

$$\sigma_y = \sqrt{\frac{\Sigma y^2}{n} - \left(\frac{\Sigma y}{n}\right)^2} = 2.345$$

$$r_{xy} = \frac{\text{COV}(x, y)}{\sigma_x \sigma_y} = \frac{3}{2.121 \times 2.345} = 0.6032$$

Ex Compute the coefficients of correlation b/w X and Y using the following data.

X	65	67	66	71	67	70	68	69
Y	67	68	68	70	64	67	72	70

Ans 0.3974

Correlation Coefficient

Let U and V are the ranks of the RVs X and Y , respectively. The correlation coefficient between U and V is called the rank correlation coeff b/w X and Y . (ρ_{xy}).

Let U and V takes the values $1, 2, \dots, n$

$$D = U - V$$

$$\rho_{xy} = 1 - \frac{6 \sum d^2}{n(n^2-1)}$$

[Spearman's formula]

Ex The marks secured by some 9 students in Maths and Physics are given below -

Maths	Physics	R_x	R_y	$d = (R_x - R_y)$	d^2
10	30	9	9	0	0
15	42	5	3	2	4
12	45	8	2	6	36
17	46	3	1	2	4
13	33	7	8	1	1
16	34	4	7	-3	9
24	40	1	4	-3	9
14	35	6	6	0	0
22	39	2	5	-3	9

$$\rho = 1 - \frac{6 \sum d^2}{n(n^2-1)} = 1 - \frac{6 \times 72}{9 \times 80} = 1 - 0.6 = 0.4$$

$\sum d^2 = 72$

If the Ranks are repeated

$$\rho = 1 - \frac{6(\sum d^2 + C)}{n(n^2 - 1)}$$

where $C = \frac{m(m^2 - 1)}{12}$
(correction factor)

(m is the freq. of the rank)

x	y	R _x	R _y	d ² = (R _x - R _y) ²
68	46 62	4	5	1
(64)	58	6	7	1
(75)	(68)	2.5	3.5	1
50	45	9	10	1
(64)	81	6	1	25
80	60	1	6	25
(75)	(68)	2.5	3.5	1
40	48	10	9	1
55	50	8	8	0
(64)	74	6	2	16

$$\sum d^2 = 72$$

$$C = 2 \frac{(4-1)}{12} + 3 \frac{(9-1)}{12} + 2 \frac{(4-1)}{12} = 3$$

$$\rho = 1 - \frac{6(72 + 3)}{10 \times 99} = 1 - \frac{6 \times 75}{990} = 0.545$$

Assumption of the Pearsonian Coefficient

- ① There is linear relationship b/w the variables.
- ② The two variables under study are affected by a large no. of independent causes so as to form a normal distribution.
- ③ There is a cause and effect relationship b/w the forces affecting the distribution of the items.

Features of Spearman's Correlation Coefficient

- ↳ It is distribution-free or non-parametric because no strict assumption are made about the form of population.

Ex The competitors in an contest are ranked by three judges. Determine which pair of judges has the nearest approach to common tastes.

1 st judge rank R_1	2 nd judge rank R_2	3 rd judge rank R_3	$(R_1 - R_2)^2$	$(R_2 - R_3)^2$	$(R_1 - R_3)^2$
1	3	6	4	9	25
6	5	4	1	1	4
5	8	9	9	1	16
10	4	8	36	16	4
3	7	1	16	36	4
2	10	2	64	64	0
4	2	3	4	1	1
9	1	10	64	81	1
7	6	5	1	1	4
8	9	7	1	4	1
$N = 10$			$\Sigma D^2 = 200$	$\Sigma D^2 = 214$	$\Sigma D^2 = 60$

Correlation b/w 1st and 2nd judges

$$R_{1-2} = 1 - \frac{6 \sum D^2}{N(N^2-1)} = 1 - \frac{6 \times 200}{10(99)} = 1 - 1.212 = -0.212$$

Similarly b/w 2nd and 3rd

$$R_{2-3} = 1 - \frac{6 \times 214}{990} = -0.292$$

B/w 1st and 3rd

$$R_{1-3} = 1 - \frac{6 \times 60}{990} = 0.636$$

∴ Judgment of 1st and 3rd judges is more correlated
 → the nearest approach to common tastes.

<u>Ex</u>	Compute Spearman's rank correlation				
	X	Y	R _x	R _y	(R _x - R _y) ² = D
	20	28	1	6	25
	22	24	2	1.5	0.25
	28	24	6	1.5	20.25
	23	25	3.5	2	0.25
	30	26	7.5	4	12.25
	30	27	7.5	5	6.25
	23	32	3.5	8	20.25
	24	30	5	7	4

$$\sum D^2 = 88.5$$

$$R = 1 - \frac{6(\sum D^2 + C)}{N(N^2-1)}$$